

Sign-positivity for $(3, 2)/(4, 1)$: a structural reduction to two multiplicity bounds

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Abstract

The sign-positivity lemma underlying Theorem B (multiset form) asserts $M_{(4,1)} \leq 2M_{(3,2)}$ and $M_{(3,2)} \leq 2M_{(4,1)}$ componentwise on B_4 . We give a structural reduction: combining the $2q$ -difference identity $A_{(3,2)} - A_{(4,1)} = 2q(q^2 + q + 1)$ with the canonical B_4 -decomposition of $2q(q^2 + q + 1)$, the sign-positivity lemma is *equivalent* to the pair of multiplicity bounds

$$m_2^{(3,2)} \geq 2 \quad \text{and} \quad m_1^{(4,1)} \geq 2.$$

This isolates the load-bearing inequalities: each of them concerns a single component of a single σ_1 -G1 atom, and each is sharp up to the slack-1 component of atom_{RTL} (resp. atom_{LTR}). The reduction also shows why the slack $\text{atom}_{\text{RTL}} - M_{(3,2)} = (0, 2, -2)$ and $\text{atom}_{\text{LTR}} - M_{(4,1)} = (0, -2, 2)$ are antipodal on the $\bar{c}$ -axis: each is the perturbation $\pm(0, 2, -2)$ corresponding to the canonical decomposition of $2q(q^2 + q + 1)$ on B_4 . We verify the two bounds computationally and discuss what a W -graph-natural proof would require.

1 Setup

We adopt without re-derivation the framework of `2026-05-06-twin-multiset.tex` and `2026-05-06-theorem-B-m` (same directory). Recall the canonical σ_1 -G1 basis

$$B_{2k} = \{q^c(1+q)^{2k-2c} : c = 0, 1, \dots, k\} \subset \mathbb{Z}[q],$$

linearly independent over $\mathbb{Q}[q]$. Each σ_1 -G1 atom $A_\lambda(q)$ admits a unique non-negative integer expansion in B_{2m_λ} where $m_\lambda = (\deg A_\lambda)/2$, and we denote its multiplicity vector by $M_\lambda = (m_0^{(\lambda)}, \dots, m_{m_\lambda}^{(\lambda)})$.

For Theorems A, B, C, the relevant atoms at $n \in \{4, 5\}$ are

$$M_{(3,1,1)} = (1, 2), \quad M_{(3,2)} = (1, 5, 3), \quad M_{(4,1)} = (1, 3, 5). \quad (1)$$

Both $M_{(3,2)}$ and $M_{(4,1)}$ are on B_4 . They are not reflections of each other under $\bar{c} \mapsto 2 - \bar{c}$ (their reflected versions would be $(3, 5, 1)$ and $(5, 3, 1)$); rather, they are individually palindromic in the polynomial sense (since A_λ is always palindromic in q).

2 Sign-positivity, multiset form

Definition 1 (Sign-positivity). The *sign-positivity lemma at B_4 for the pair $(3, 2)/(4, 1)$* is the assertion that the two signed differences

$$2M_{(3,2)} - M_{(4,1)} \quad \text{and} \quad 2M_{(4,1)} - M_{(3,2)}$$

are non-negative as integer-valued functions on $\{0, 1, 2\}$. We denote them

$$\begin{aligned}\text{atom}_{\text{RTL}} &:= 2M_{(3,2)} - M_{(4,1)}, \\ \text{atom}_{\text{LTR}} &:= 2M_{(4,1)} - M_{(3,2)}.\end{aligned}$$

The sign-positivity lemma is the non-trivial input to Theorem B's multiset form (proved May 6 in 2026-05-06-theorem-B-multiset.tex): convolving by $M_{(3,1,1)} = (1, 2)$ promotes the two atoms to $M_{(3,2,1)}$ and $M_{(5,1)}$ respectively. By Proposition 3 of that writeup,

$$M_{(3,1,1)} * \text{atom}_{\text{RTL}} = M_{(3,2,1)}, \quad M_{(3,1,1)} * \text{atom}_{\text{LTR}} = M_{(5,1)}.$$

We now show that the sign-positivity lemma admits a clean structural reduction to two specific multiplicity bounds.

3 The $2q$ -difference and its canonical decomposition

Lemma 2 ($2q$ -difference, restated from 2026-05-06-theorem-B-multiset.tex).

$$A_{(3,2)}(q) - A_{(4,1)}(q) = 2q(q^2 + q + 1).$$

Proof. By direct expansion of the multiplicity vectors on B_4 :

$$A_{(3,2)} - A_{(4,1)} = [(1+q)^4 + 5q(1+q)^2 + 3q^2] - [(1+q)^4 + 3q(1+q)^2 + 5q^2] = 2q(1+q)^2 - 2q^2.$$

$$\text{And } 2q(1+q)^2 - 2q^2 = 2q[(1+q)^2 - q] = 2q(q^2 + q + 1). \quad \square$$

Lemma 3 (Canonical B_4 -decomposition of $2q(q^2 + q + 1)$). *The polynomial $2q(q^2 + q + 1)$ has the unique B_4 -decomposition*

$$2q(q^2 + q + 1) = 0 \cdot (1+q)^4 + 2 \cdot q(1+q)^2 + (-2) \cdot q^2.$$

Equivalently, viewed as a signed multiplicity vector on $\{0, 1, 2\}$:

$$M_{2q(q^2+q+1)} = (0, +2, -2).$$

Proof. Expanding: $0 \cdot (1+q)^4 + 2 \cdot q(1+q)^2 + (-2) \cdot q^2 = 2(q + 2q^2 + q^3) - 2q^2 = 2q + 2q^2 + 2q^3 = 2q(q^2 + q + 1)$. Uniqueness is the linear independence of B_4 over $\mathbb{Q}[q]$. $\square$

Corollary 4 (Multiplicity-vector form of the $2q$ -difference).

$$M_{(3,2)} - M_{(4,1)} = (0, +2, -2).$$

Equivalently, $M_{(3,2)}$ has +2 paths at $\bar{c} = 1$ and -2 paths at $\bar{c} = 2$ relative to $M_{(4,1)}$, with equal counts at $\bar{c} = 0$.

Proof. Combine Lemmas 2 and 3: $M_{(3,2)} - M_{(4,1)} = M_{A_{(3,2)} - A_{(4,1)}} = M_{2q(q^2+q+1)} = (0, +2, -2)$. $\square$

This corollary shows the differences componentwise: $(1, 5, 3) - (1, 3, 5) = (0, +2, -2)$. The point of stating it this way is to set up the structural reduction.

4 Structural reduction of sign-positivity

We now reduce the sign-positivity lemma to two specific multiplicity bounds.

Theorem 5 (Structural sign-positivity reduction). *The sign-positivity lemma ($\text{atom}_{\text{RTL}} \geq 0$) and ($\text{atom}_{\text{LTR}} \geq 0$) on $\{0, 1, 2\}$ is equivalent to the conjunction of the two multiplicity bounds*

$$\boxed{m_2^{(3,2)} \geq 2} \quad \text{and} \quad \boxed{m_1^{(4,1)} \geq 2}.$$

Proof. By Corollary 4, $M_{(3,2)} = M_{(4,1)} + (0, +2, -2)$ and $M_{(4,1)} = M_{(3,2)} + (0, -2, +2)$. We treat each of the two atoms in turn.

Computing atom_{RTL} . By definition,

$$\begin{aligned} \text{atom}_{\text{RTL}} &= 2M_{(3,2)} - M_{(4,1)} \\ &= M_{(3,2)} + (M_{(3,2)} - M_{(4,1)}) \\ &= M_{(3,2)} + (0, +2, -2) \\ &= (m_0^{(3,2)}, m_1^{(3,2)} + 2, m_2^{(3,2)} - 2). \end{aligned}$$

The non-negativity $\text{atom}_{\text{RTL}} \geq 0$ is therefore the conjunction of the three inequalities $m_0^{(3,2)} \geq 0$ (trivial), $m_1^{(3,2)} + 2 \geq 0$ (trivial since $m \geq 0$), and

$$m_2^{(3,2)} \geq 2.$$

The first two are automatic; the third is a non-trivial multiplicity bound.

Computing atom_{LTR} . Symmetrically,

$$\begin{aligned} \text{atom}_{\text{LTR}} &= 2M_{(4,1)} - M_{(3,2)} \\ &= M_{(4,1)} + (M_{(4,1)} - M_{(3,2)}) \\ &= M_{(4,1)} + (0, -2, +2) \\ &= (m_0^{(4,1)}, m_1^{(4,1)} - 2, m_2^{(4,1)} + 2). \end{aligned}$$

The non-negativity $\text{atom}_{\text{LTR}} \geq 0$ is therefore the conjunction of $m_0^{(4,1)} \geq 0$ (trivial), $m_2^{(4,1)} + 2 \geq 0$ (trivial), and

$$m_1^{(4,1)} \geq 2.$$

So the sign-positivity lemma is equivalent to the two non-trivial bounds $m_2^{(3,2)} \geq 2$ and $m_1^{(4,1)} \geq 2$. $\square$

Remark 6 (Why this is structural). Theorem 5 is a real structural reduction in the following sense. Each of the two bounds concerns a *single component* of a *single* σ_1 -G1 atom, and each is independent of any cross-comparison between the two cells. So the proof of sign-positivity — and through it, the multiset form of Theorem B — reduces to the two specific path-counting bounds.

The bounds are sharp up to the slack at the corresponding component:

$$m_2^{(3,2)} - 2 = \text{atom}_{\text{RTL},2} = 1, \quad m_1^{(4,1)} - 2 = \text{atom}_{\text{LTR},1} = 1.$$

So the two bounds saturate at exactly the slack of the corresponding atom-RTL/LTR multiplicity at the sole non-palindromic-component-of-the-difference. (The other two components have automatic slack.)

5 Verification of the two bounds

Lemma 7 (Computational verification of the bounds).

$$m_2^{(3,2)} = 3 \geq 2 \quad \text{and} \quad m_1^{(4,1)} = 3 \geq 2.$$

Proof. By direct enumeration of closed W -graph paths in $V_{(3,2)}$ and $V_{(4,1)}$ at S_5 .

The W -graph of $V_{(3,2)} \subset S_5$ has 5 vertices (SYTs of shape $(3,2)$) and 6 edges. Closed paths along the staircase reduced word for $w_0 \in S_5$ (length 10) at post-prefix bigrade $\bar{c} = 2$ have $|S| = 2$. Direct enumeration (`~/projects/scratch/2026-05-06-paths/path_enumerate.py`, May 6, 2026) gives exactly 3 such paths, each with $|S| = 2$ and starting from one of two distinguished vertices (vertex 1 with multiplicity 2 and vertex 2 with multiplicity 1).

The W -graph of $V_{(4,1)} \subset S_5$ has 4 vertices and 3 edges (a path graph P_4). Closed paths at post-prefix bigrade $\bar{c} = 1$ have $|S| = 6$. Direct enumeration gives exactly 3 such paths, each starting from vertex 1.

In particular both $m_2^{(3,2)} = 3$ and $m_1^{(4,1)} = 3$, so the bounds ≥ 2 hold strictly with slack 1 each. $\square$

Corollary 8 (Sign-positivity, proved). *The sign-positivity lemma holds: $\text{atom}_{\text{RTL}} = (1, 7, 1) \geq 0$ and $\text{atom}_{\text{LTR}} = (1, 1, 7) \geq 0$ as functions on $\{0, 1, 2\}$.*

Proof. Combine Theorem 5 with Lemma 7. $\square$

6 Why the reduction is interesting

The slack lives in a single component

The vector $\text{atom}_{\text{RTL}} - M_{(3,2)} = (0, +2, -2)$ has its non-trivial structure at $\bar{c} \in \{1, 2\}$: $V_{(3,2)}$ has 2 “extra” paths at $\bar{c} = 1$ relative to $V_{(4,1)}$, and 2 “missing” paths at $\bar{c} = 2$. The bound $m_2^{(3,2)} \geq 2$ exactly absorbs the missing-2 at $\bar{c} = 2$.

Symmetrically, $\text{atom}_{\text{LTR}} - M_{(4,1)} = (0, -2, +2)$, and $m_1^{(4,1)} \geq 2$ absorbs the missing-2 at $\bar{c} = 1$ for the symmetric atom.

Notice the antipodal pattern: the extra-2 / missing-2 swap between $\bar{c} = 1$ and $\bar{c} = 2$ as we swap $(3, 2)$ and $(4, 1)$. This is a *statement* about the $2q(q^2 + q + 1)$ -form of the difference, not about any individual cell.

What the bounds mean structurally

The bound $m_2^{(3,2)} \geq 2$ says: there are at least 2 closed W -graph paths in $V_{(3,2)}$ at S_5 at $\bar{c} = 2$ (i.e., with $|S| = 2$, the maximum possible $|S^c|$ within the post-prefix range). The actual count is 3, so the bound has slack 1.

The bound $m_1^{(4,1)} \geq 2$ similarly says: there are at least 2 closed W -graph paths in $V_{(4,1)}$ at S_5 at $\bar{c} = 1$. The actual count is 3.

What is missing

Theorem 5 reduces sign-positivity to two specific path-counting bounds. We verified them computationally (Lemma 7). What we have *not* done is give a W -graph-structural proof that does not invoke direct enumeration. Such a proof would presumably use:

- For $m_2^{(3,2)} \geq 2$: the structure of the $V_{(3,2)}$ W -graph (which is the bipartite graph $K_{2,3}$, with parts $\{1, 4\}$ and $\{0, 2, 3\}$ in our enumeration). Closed walks of length 8 in $K_{2,3}$ are abundant; a direct lower bound from the spectrum of the adjacency matrix gives much more than 2. The non-trivial part is to count closed walks satisfying the W -graph D -step constraint at $|S| = 2$.
- For $m_1^{(4,1)} \geq 2$: the structure of the $V_{(4,1)}$ W -graph (path graph P_4). Closed walks at $|S| = 6$ from vertex 1 (the unique starting vertex by the data) are constrained by the linear topology.

Even for these tiny cases, a structural lower bound bypassing enumeration is non-trivial. The natural framework would be a transfer-operator estimate on the W -graph, where each D -step contributes $1+q$ and each M -step contributes v , and one bounds the closed-walk generating function by the spectral radius of an appropriate operator. We do not pursue this here.

Comparison with the path-level Theorem B

The path-level Theorem B (proved May 6 in 2026-05-06-theorem-B-paths.tex by pigeonhole and explicitly tabulated in 2026-05-06-paths-explicit-and-Q3-refuted.tex) gives an existence-and-explicit injection of paths with the right slack. The current theorem reduces the sign-positivity *numerical* content to two single-component path-count bounds.

The two perspectives complement each other:

- Theorem B (path-level): asserts the existence of an injection of multisets of paths.
- Theorem 5: identifies the load-bearing inequalities as $m_2^{(3,2)} \geq 2$ and $m_1^{(4,1)} \geq 2$.

A canonical path-level injection would simultaneously verify both inequalities via two specific injections of small finite sets, with the `atomRTL` slack identified as a specific extra path at $\bar{c} = 2$. Pinning down which specific path is the slack remains open (Section 7).

7 What this does not give: canonical injection

Theorem 5 reduces sign-positivity to two integer inequalities. It does *not* pin down the specific injection of multisets that achieves the inequality. The lex injection of 2026-05-06-paths-explicit-and-Q3-refuted. is one specific injection; many others work equally well.

A canonical injection would identify, structurally, which path in $V_{(3,2)}$ at $\bar{c} = 2$ is “the slack” — equivalently, which 2 of the 3 paths are “imaged from $V_{(4,1)}$ ” and which 1 is “new” to $V_{(3,2)}$.

We note one structural observation that does not, however, pin down the canonical injection:

Remark 9 (Branching component of the slack at $\bar{c} = 2$). At $\bar{c} = 2$, the three paths in $V_{(3,2)}$ split by branching component as

- Two paths with starting vertex in $V_{(3,1)}$ -branching (vertex 1, lying in the $V_{(3,1)}$ -summand of $V_{(3,2)} \downarrow S_4$).
- One path with starting vertex in $V_{(2,2)}$ -branching (vertex 2, lying in the $V_{(2,2)}$ -summand).

Meanwhile, all five paths in $V_{(4,1)}$ at $\bar{c} = 2$ start at a $V_{(3,1)}$ -branching vertex (vertex 2 of $V_{(4,1)}$, lying in the $V_{(3,1)}$ -summand of $V_{(4,1)} \downarrow S_4$).

If the canonical injection respected the branching identification $V_{(3,2)} \downarrow V_{(3,1)} \leftrightarrow V_{(4,1)} \downarrow V_{(3,1)}$ on starting vertices, we would inject 5 $V_{(4,1)}$ -paths into 2 (paths) $\times$ 2 (copies) = 4 slots — one short. So at least one $V_{(4,1)}$ -path must map to the $V_{(2,2)}$ -branching slot.

This is a *lower bound* on the non-canonical-by-branching content of the injection at $\bar{c} = 2$, but it does not single out a specific path.

The corresponding observation at $\bar{c} = 0$ (where the slack is also 1) does not have the same flavor: the unique $V_{(3,2)}$ path at $\bar{c} = 0$ has starting vertex 0 (in the $V_{(3,1)}$ -branching component of $V_{(3,2)}$), and the unique $V_{(4,1)}$ path at $\bar{c} = 0$ has starting vertex 0 (in the $V_{(4)}$ -branching component of $V_{(4,1)}$). These are in different branching summands, but neither is forced to be the “slack” since the slack at $\bar{c} = 0$ is the $V_{(3,2)}$ path in copy 2 (i.e., a copy-choice rather than a path-choice).

8 Open questions

1. **Structural lower bounds for the two multiplicity bounds.** Give a W -graph-natural proof that $m_2^{(3,2)} \geq 2$ and $m_1^{(4,1)} \geq 2$, without enumerating all closed paths.
2. **Canonical injection.** Pin down a W -graph-natural injection $\iota : \text{Paths}(V_{(4,1)}) \hookrightarrow \text{Paths}(V_{(3,2)}) \sqcup \text{Paths}(V_{(3,2)})$ preserving the bigrading, with the slack identified as a specific structural feature.
3. **Generalization.** Does a similar reduction hold for other partition pairs (λ, μ) at S_n with $A_\lambda - A_\mu$ in a similar simple form? Specifically, when $A_\lambda - A_\mu = N \cdot p(q)$ for some palindromic $p(q)$ and integer N , is the sign-positivity lemma $|NM_p| \leq M_\lambda$ reducible to a small number of multiplicity bounds, with the slack absorbed in a few specific components?

Computational summary

All computations performed in `~/projects/scratch/2026-05-06-sign-positivity/` and `~/projects/scratch/`
Cross-checks:

- Multiplicity vectors $M_{(3,2)} = (1, 5, 3)$, $M_{(4,1)} = (1, 3, 5)$ (Apr 24, recomputed May 6).
- $A_{(3,2)} - A_{(4,1)} = 2q(q^2 + q + 1)$ (Apr 29 polynomial residual zero).
- B_4 -decomposition $2q(q^2 + q + 1) = 0 + 2q(1 + q)^2 - 2q^2$ (Lemma 3).
- Path counts $|\text{Paths}(V_{(3,2)})_{\bar{c}}| = (1, 5, 3)$ and $|\text{Paths}(V_{(4,1)})_{\bar{c}}| = (1, 3, 5)$ at S_5 (May 6, `path_enumerate.py`).
- Sign-positivity numerical: $\text{atom}_{\text{RTL}} = (1, 7, 1)$ and $\text{atom}_{\text{LTR}} = (1, 1, 7)$, both non-negative (May 6, `test_theorem.B.py`).